AQ10.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
Level 1:

Consider a function $f: R^2 \to R$ defined as:

$$f(x, y) = \begin{cases} x^2 \sin\frac{1}{x} + y & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

- **Statement 1:** f_y is continuous at the origin.
- **Statement 2:** f_x is continuous at the origin.

Enter the number of correct statements.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

1 point
A function $f: R^2 \to R$ satisfies that $f_x(x, y) = c$ and $f_y(x, y) = k$ of all points of R^2, where c and k are some constants, then which of the following option(s) is (are) true?

☐ The tangent line exists at any point of R^2 in any direction.
☐ The tangent lines at the origin do not lie on a plane.
☐ There exist points in R^2 for which tangent points in all directions do not exist.
☐ $f(x, y)$ can be $f(x, y) = cx + ky + \alpha$, where α is any real number.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The tangent line exists at any point of R^2 in any direction.
$f(x, y)$ can be $f(x, y) = cx + ky + \alpha$, where α is any real number.

1 point
Consider $f: R^2 \to R$ and suppose partial derivatives of $f(x, y)$ with respect to x and y are continuous at a point (a, b). Then which of the following option(s) is (are) true?

☐ $f_x(a, b) = \lim_{(x,y) \to (a,b)} f_x(x, y)$

☐ $f_y(a, b) = \lim_{(x,y) \to (a,b)} f_y(x, y)$

☐ $f_x(a, b) \neq \lim_{(x,y) \to (a,b)} f_x(x, y)$.

☐ $f_y(a, b) \neq \lim_{(x,y) \to (a,b)} f_y(x, y)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$f_x(a,b) = \lim_{(x,y)\to(a,b)} f_x(x,y)$fx(a, b) = (x,y)→(a,b)lim fx(x, y)

$f_y(a,b) = \lim_{(x,y)\to(a,b)} f_y(x,y)$fy(a, b) = (x,y)→(a,b)lim fy(x, y)

1 point

Consider a function $f(x,y) = x^2 + y^2$ f(x, y) = x2 + y2. Which of following is the tangent at the point (1,2) in the direction of (1, 1)?

- ○ $x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$x = 1 + √2 t, y(t) = 2 + √2 t, z(t) = 5 + √2 6t
- ○ $x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t$x = 1 + t, y(t) = 2 + t, z(t) = 5 + 6t
- ○ $x = 1, y(t) = 2 + t, z(t) = 5 + \frac{6t}{\sqrt{2}}$x = 1, y(t) = 2 + t, z(t) = 5 + √2 6t
- ○ $x = 1 + t, y(t) = 2 + t, z(t) = 4 + \frac{6t}{\sqrt{2}}$x = 1 + t, y(t) = 2 + t, z(t) = 4 + √2 6t

No, the answer is incorrect.
Score: 0
Accepted Answers:

$x = 1 + \frac{t}{\sqrt{2}}, y(t) = 2 + \frac{t}{\sqrt{2}}, z(t) = 5 + \frac{6t}{\sqrt{2}}$x = 1 + √2 t, y(t) = 2 + √2 t, z(t) = 5 + √2 6t

Level 2:

1 point

Consider a function

$$f(x,y) = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ 0 & \text{if } y = 0. \end{cases}$$

.f(x, y) = {yx0if y = 0if y = 0..

Which of the following is the tangent line passing through the point (1, -1) and parallel to the line

$x(t) = \frac{t}{\sqrt{5}}, y(t) = 1 + \frac{2t}{\sqrt{5}}, z(t) = -\frac{3t}{\sqrt{5}}$x(t) = 5
√
t, y(t) = 1 + 5
√
2t, z(t) = − 5
√
3t?

○
$x(t) = 1 + t, y(t) = -1 + 2t, z(t) = -\frac{3}{\sqrt{5}}t$x(t) = 1 + t, y(t) = −1 + 2t, z(t) = − 5
√
3t

○
$x(t) = 1 + t, y(t) = 1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$x(t) = 1 + t, y(t) = 1 + 2t, z(t) = −1 − 5
√
3t

○
$x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$x(t) = 1 + 5
√
t, y(t) = −1 + 5
√
2t, z(t) = −1 − 5
√
3t

○
$x(t) = -1 + t, y(t) = -1 + 2t, z(t) = -1 - \frac{3}{\sqrt{5}}t$x(t) = −1 + t, y(t) = −1 + 2t, z(t) = −1 − 5
√
3t

No, the answer is incorrect.
Score: 0
Accepted Answers:

$x(t) = 1 + \frac{t}{\sqrt{5}}, y(t) = -1 + \frac{2t}{\sqrt{5}}, z(t) = -1 - \frac{3t}{\sqrt{5}}$x(t) = 1 + 5
√
t, y(t) = −1 + 5
√
2t, z(t) = −1 − 5
√
3t

1 point

Consider a function $f(x, y, z) = 2x^2 + xy + z^2$ f(x, y, z) = 2x2 + xy + z2. Which of following is the tangent at the point (1,1,1) in the direction of (1, 0, 0)

○
$(1, 1, 1, 4 + 5t)$(1, 1, 1, 4 + 5t)

○
$(1 + t, 1, 1, 4 + t)$(1 + t, 1, 1, 4 + t)

○
$(1, 1 + t, 1 + t, 4 + 5t)$(1, 1 + t, 1 + t, 4 + 5t)

○

$(1 + t, 1, 1, 4 + 5t)$(1 + t, 1, 1, 4 + 5t)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(1 + t, 1, 1, 4 + 5t)$(1 + t, 1, 1, 4 + 5t)

1 point

Consider the following functions ff defined as:

$$f(x, y, z) = xyz.$$f(x, y, z) = xyz.

Then the tangent line at the point (1,0,1) in the direction of (1,1,1) is parallel to the line

○ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3 √ t, y(t) = 3 √ t, z(t) = 1 + 3 √ t, u(t) = 3 √ t

○ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = t\sqrt{3}$x(t) = 1 + 3 √ t, y(t) = 3 √ t, z(t) = 1 + 3 √ t, u(t) = t 3 √

○ $x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = 1 + \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3 √ t, y(t) = 1 + 3 √ t, z(t) = 1 + 3 √ t, u(t) = 3 √ t

○ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$x(t) = 1 + \frac{t}{\sqrt{3}}, y(t) = \frac{t}{\sqrt{3}}, z(t) = 1 + \frac{t}{\sqrt{3}}, u(t) = \frac{t}{\sqrt{3}}$x(t) = 1 + 3

$\sqrt{}$

$t, y(t) = \quad 3$

$\sqrt{}$

$t, z(t) = 1 + \quad 3$

$\sqrt{}$

$t, u(t) = \quad 3$

$\sqrt{}$

t

Check Answers

Your score is: 0/7